

Project — Online Quiz & Exam Platform

Overview

A multi-role online learning and assessment platform. Students enrol in courses and take quizzes with live scoring and performance tracking. Instructors build course content and analyse results. Teaching Assistants support course delivery by creating practice material, monitoring struggling students, and managing doubt sessions. The platform admin governs users, courses, and academic integrity across the institution.

Roles

Role	Responsibility
Student	Enrolment, quiz-taking, results, performance tracking

Role	Responsibility
Instructor	Course and quiz creation, grading analytics, materials
Teaching Assistant	Practice quizzes, student monitoring, doubt sessions

Role	Responsibility
Admin	User management, platform settings, institutional reports

Shared Database Schema

All four students create and use the same database. The schema must include at minimum the following tables:

users – All platform users

- `id`, `name`, `email`, `password_hash`, `phone`, `role` (student/instructor/ta/admin), `profile_pic`, `student_id`, `program`, `is_active`, `created_at`

subjects – Top-level academic subjects (e.g., Mathematics, Computer Science)

- `id`, `name`, `description`

courses – Courses created by instructors

- `id`, `instructor_id`, `subject_id`, `title`, `description`, `enrollment_type` (open/approval), `max_students`, `status` (draft/active/archived), `created_at`

course_tas – Teaching assistants assigned to a course

- `id`, `course_id`, `ta_id`, `assigned_at`

enrollments – Student enrolment in courses

- `id`, `student_id`, `course_id`, `status` (pending/active/dropped), `enrolled_at`

quizzes – Quizzes and practice tests within a course

- `id, course_id, created_by, title, description, time_limit_minutes, total_marks, pass_mark, quiz_type (graded/practice), status (draft/published), available_from, available_until`

`questions` – MCQ questions belonging to quizzes

- `id, quiz_id, question_text, marks, order_index, created_by`

`options` – Answer options for each question

- `id, question_id, option_text, is_correct`

`attempts` – Student quiz attempts

- `id, quiz_id, student_id, score, started_at, completed_at, is_graded`

`answers` – Individual answers within an attempt

- `id, attempt_id, question_id, selected_option_id`

`course_materials` – Files and links uploaded for a course

- `id, course_id, uploaded_by, title, file_path, material_type (document/link/video), created_at`

`announcements` – Course-level announcements

- `id, course_id, author_id, title, body, created_at`

`qa_questions` – Student questions in the course Q&A board

- `id, course_id, student_id, title, body, is_resolved, created_at`

`qa_answers` – Replies to Q&A board questions

- `id, qa_question_id, author_id, body, is_endorsed, created_at`

`doubt_sessions` – TA-scheduled live doubt sessions

- `id, course_id, ta_id, title, scheduled_at, duration_minutes, location_or_link, max_attendees`

`doubt_session_bookings` – Students who have booked a doubt session slot

- `id, doubt_session_id, student_id, booked_at`

Each group member is responsible for creating their own data in the database. Each member may only insert the data that is required for demonstrating their own work

Technical Requirements

- Follow the **MVC pattern**: separate Models, Views, and Controllers; no business logic in view files
- Use **PHP** for all server-side logic and **MySQL** as the database
- All database queries must use **mysqli with prepared statements** — no raw string insertion allowed
- Authentication must use **PHP sessions** with role-based access control on every protected page

- Each student must implement **at least one feature using AJAX** (XMLHttpRequest) communicating with a PHP API endpoint that returns JSON
- Maintain a central repository for the group where each student will push their code. It is the responsibility of each group member to ensure their code works when cloned and run in a XAMPP Apache server.
- Use **Git** on the command line for version control. Do not push to main/master branch directly. Create feature branches and submit a Pull Request for each major feature
- Submit: (1) working online codebase, (2) hardcopy report describing your role's features

Role 1 — Student

Description

Students are the learners on the platform. They enrol in courses, consume study materials, interact with the Q&A board, take graded and practice quizzes, review their results, and track their academic progress over time.

Features

- Register with name, email, student ID, program, and password; log in and log out
- Manage profile: update personal information, upload profile picture, change password
- Browse all active courses by subject; search by keyword; view course description, instructor name, and enrolled student count
- Enrol in open-enrollment courses directly; submit an enrolment request for approval-based courses
- View all enrolled courses on a personal dashboard with course status and next upcoming quiz
- View enrolled course detail: description, instructor, assigned TA, announcements, materials, and quizzes
- Access and download course materials (documents, links, videos) uploaded by the instructor or TA
- View and read all course announcements
- Browse the course Q&A board; post a new question (title + body); view all questions and their answers
- Mark own Q&A question as resolved; view questions answered by instructors or TAs with endorsement badges
- View all published graded quizzes for enrolled courses; take a quiz (MCQ, single page, countdown timer)
- View quiz result immediately after submission: score, pass/fail, and question-by-question breakdown
- Take unlimited attempts on practice quizzes; view results after each attempt
- View complete attempt history for all quizzes with scores and timestamps

- View course leaderboard (top students by score for a specific quiz) via AJAX
- View personal performance dashboard: scores over time, pass rate, average score per subject, and comparison to class average
- Book a slot in a TA's doubt session; view all upcoming booked doubt sessions
- Drop a course (if no graded quiz has been completed in it)

Role 2 — Instructor

Description

Instructors create and manage courses, build quiz content, upload learning resources, and track the academic performance of their students. They engage students through the Q&A board and course announcements.

Features

- Log in (account created by admin); manage professional profile: name, department, bio, profile picture
- Create a new course: title, subject, description, enrollment type (open/approval), and maximum student count; save as draft or publish
- Manage course details: edit all fields, archive a completed course; view enrolled student list
- Manage course enrollment requests: approve or reject pending enrollment requests
- Assign a Teaching Assistant to a course from the list of TA accounts
- Create quizzes for their course: title, description, time limit (minutes), total marks, pass mark, type (graded/practice), and availability window (from date to until date)
- Add MCQ questions to a quiz: question text, marks, and four answer options with the correct one marked
- Manage a course-level question bank: edit, delete, or reuse questions across multiple quizzes
- Publish or unpublish quizzes; view all quizzes with attempt counts and average scores
- View all student attempts for a specific quiz: student name, score, duration, and pass/fail
- View grade analytics for each quiz: class average, highest and lowest score, score distribution, pass rate
- Post course announcements: title and body; view all past announcements
- Upload course materials: title, file upload or external link, and material type
- Manage uploaded materials: edit title, replace file, or delete
- Answer and endorse answers on the course Q&A board; mark Q&A questions as resolved
- View overall course performance report: enrolled students, quiz completion rates, average scores per quiz, drop-out rate

Role 3 — Teaching Assistant

Description

Teaching Assistants support the instructor in delivering the course. They create practice quizzes, monitor student progress to identify those who are struggling, manage the course Q&A board, upload supplementary resources, and schedule doubt sessions for students.

Features

- Log in and manage profile; view all courses they are assigned to
- View course details, enrolled students, and all quizzes for each assigned course
- Create practice quizzes for an assigned course (quiz type = practice, requires instructor approval before appearing to students)
- Manage question bank for assigned courses: add, edit, and delete questions and options
- View all student attempt results for their assigned courses: individual scores, timestamps, and pass/fail status
- Identify and flag at-risk students: view a filtered list of students whose quiz scores fall below a configurable threshold; flag them for instructor review
- Send a course announcement on behalf of the instructor (marked as "From TA")
- Upload supplementary study materials for an assigned course: notes, cheat-sheets, additional references
- Manage uploaded materials: edit or delete their own uploads
- Monitor and respond to student questions on the course Q&A board
- Endorse particularly helpful answers posted by students or other TAs
- Schedule doubt sessions: create a session with title, date/time, duration, location or meeting link, and maximum attendees
- View all bookings for each doubt session; view list of attending students
- Cancel or reschedule a doubt session with a notice to all booked students
- View a summary report for each assigned course: total students, quiz attempt rates, average score, number of at-risk students

Role 4 — Admin

Description

The Admin manages the entire platform: user accounts, course catalog oversight, academic integrity enforcement, and institutional reporting. They are the only user who can approve instructor accounts and promote users to TA or admin roles.

Features

- Log in to an admin dashboard: total users (by role), total active courses, total quiz attempts today, and total pending instructor approval requests
- Manage all user accounts: search, view, activate, deactivate, and change roles for students, instructors, TAs, and other admins
- Approve or reject new instructor registration requests; create TA accounts directly
- View all courses across the platform: active, draft, and archived; filter by subject and instructor
- Manage platform-wide subject taxonomy: add, rename, and delete subjects
- View all quizzes across the platform: filter by course, status, or type; view attempt counts
- Handle reported content or academic integrity flags: view reports, review details, and mark as resolved or escalated
- View platform-wide analytics: total enrollments per subject, quiz pass rates across all courses, most active instructors, peak usage times
- Manage platform announcements visible to all users
- View a student academic report: per-student summary of all courses, attempt counts, and average scores (searchable by student ID or name)
- Generate institutional report: total users, active courses, quizzes, attempts, and pass rates per subject for a selected semester range
- Set platform-wide policies: maximum quiz duration, maximum students per course default, and other configurable parameters
- View audit log of significant admin actions

Submission Checklist

- Each group member is responsible that each of their roles are individually accessible with their own login and dashboard
- Role-based access control prevents any user from accessing another role's pages
- The shared database schema is implemented correctly and consistently across all roles
- At least one AJAX-based feature is implemented per role
- All forms include server-side validation with descriptive error messages
- Git history shows feature branches and pull requests for major features
- Hardcopy report describes all features for your assigned role

Separation of Concerns

- Each group member is responsible for their own code and submission. Other members work will not affect one's development and submission.
- DO NOT rely on any other of your group member for anything, including DB table creation, data insertion, session management etc.